

LexiPath

Sequence-Derived Test Case Tables

Unit • Bottom-Up Integration • System End-to-End • Robustness

UC1–UC9 | Source of truth: corrected Mermaid sequence diagrams

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1. Testing Strategy and Conventions

These tables are derived strictly from sequence_diagrams_mermaid.txt. Sequence-diagram expectations remain authoritative when code or tests differ. Missing and partial automated coverage are reported separately.

Unit tests move from UI inward; bottom-up integration starts at models/services and adds HTTP routes; system tests exercise user-visible flows; robustness tests inject validation, dependency and state-integrity faults.

Coverage status legend

Status	Meaning
Implemented	A current automated test directly asserts the behavior.
Partially implemented	Related coverage exists but omits a branch, boundary or real integration.
Not implemented	No matching automated test was found.
Conflicts with sequence diagram	A current test expects different behavior.

Use synthetic records/scans only. Preserve child-written text exactly. Never expose raw scan paths. Mock Gemini, Azure and Resend unless explicitly testing a live boundary.

2. Unit Test Tables

Each table isolates one component. Mock columns define dependencies outside the unit boundary.

UC1 — Upload Student Writing Sample for Analysis

Primary endpoint: POST /api/samples/:studentId

Current automated coverage — Implemented: 3; Not implemented: 5.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC1-01 UploadSamplePage: valid submit	studentId; title; taskType; valid files	Calls uploadSample and enters analysing phase	uploadSample(FormData)	202 Sample with status UPLOADED	None	None
UT-UC1-02 UploadSamplePage: rejected upload	Invalid file or unknown student	Shows inline error; retains entered form	uploadSample(FormData)	Reject with 400/422 error	None	None
UT-UC1-03 useSamplePolling: terminal outcomes	sampleId; repeated statuses	Completes on ANALYSED/REVIEWED; fails on FAILED; times out after 30 attempts	getSample(sampleId)	Status sequence	None	None
UT-UC1-04 createSample: accepted upload	req.files; studentId; taskType	Creates Sample; responds 202; starts runAnalysis after response	Student.findById; Sample.create; runAnalysis	student; created Sample; resolved job	Samples = []	Samples = [UPLOADED sample]
UT-UC1-05 createSample: invalid student/file	Unknown student or invalid second file	Returns 400/422; cleans temporary files; creates no Sample	Student.findById; toPageFiles; fs.unlink	null or validation error; unlink succeeds	Samples = []	Samples = []
UT-UC1-06 analyseSample: valid structured result	Sample pages and task context	Returns validated errors; preserves written text; uncertainty stays confidence-based	Gemini generateContent	Valid structured JSON	Sample = UPLOADED	Sample unchanged by pure analysis call
UT-UC1-07 Gemini retry and validation	Transient failure; malformed JSON/schema; timeout	Retries to configured limit; throws final controlled failure	Gemini generateContent	Failure then success, or repeated failure	Sample = UPLOADED	Sample unchanged by callModelWithRetry
UT-UC1-08 runAnalysis: persist outcome	sampleId	Saves ANALYSED with errors or FAILED with safe analysisError	Sample.findById; analyseSample; sample.save	Sample; success or failure; saved Sample	Sample.status = UPLOADED	Sample.status = ANALYSED or FAILED

UC2 — View Categorised Error Pattern Report

Primary endpoint: GET /api/samples/:sampleId

Current automated coverage — Implemented: 3; Not implemented: 4.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC2-01 SampleReportPage: group report	Serialized Sample with live and dismissed errors	Groups live errors in category order; omits empty categories	getSample; getStudent; getLatestRecommendations	Sample; student; null	None	None
UT-UC2-02 ErrorCard: uncertainty state	confidenceScore below threshold	Renders Uncertain and confirms through confidence update	onConfirm	callback captured	None	None
UT-UC2-03 ScanViewer: page overlays	Multiple pages; indexed boxes	Shows current-page boxes; switches pages; returns errorIndex	sampleImageUrl	Synthetic image URL	None	None
UT-UC2-04 getSample: found	sampleId	Returns toClientSample with statistics and no private imagePath	Sample.findById	Sample document	Sample exists	No state change
UT-UC2-05 getSample: missing	Unknown sampleId	Returns 404 Sample not found	Sample.findById	null	Sample absent	Sample absent
UT-UC2-06 getImages: valid and invalid index	sampleId; page index	Valid index sends bytes; invalid index returns 404; path never enters JSON	Sample.findById; res.sendFile	Sample with private path; sendFile result	Sample exists	No state change
UT-UC2-07 toClientSample: statistics	Sample with dismissed/live/unsure errors	Counts non-dismissed errors; keeps unsure in total; exposes confidenceThreshold	None	None	Sample exists	No state change

UC3 — View Recommended Intervention Strategies with Rationale

Primary endpoint: GET /api/students/:studentId/
recommendations/latest

Current automated coverage — Implemented: 1; Partially implemented: 2; Not implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC3-01 RecommendationsPage: existing report	studentId; current report	Displays strategies, evidence and rationale without generation	getLatestRecommendations; getStudent; getStudentTrends	report; student; trends	None	None
UT-UC3-02 RecommendationsPage: no report	studentId; latest returns null	Shows Generate recommendations or Upload sample state	getLatestRecommendations	null	None	None
UT-UC3-03 RecommendationsPage: outdated report	report.isOutdated = true	Shows Refresh now warning; does not auto-generate	getLatestRecommendations; generateRecommendations spy	outdated report; unused spy	None	None
UT-UC3-04 getLatest: current report	Valid studentId	Returns 200 report with isOutdated false; performs no write	RecommendationReport. findOne; Sample.exists	report; null	Report exists	Report unchanged
UT-UC3-05 getLatest: stale evidence	Valid studentId; changed/new Sample	Returns same report with computed isOutdated true	RecommendationReport. findOne; Sample.exists	report; matching Sample	Report exists	Report unchanged
UT-UC3-06 getLatest: invalid/missing report	Malformed ID or no report	Returns 400 INVALID_ID or 404 RECOMMENDATION_NOT_FOUND	RecommendationReport. findOne	not called or null	Report absent	Report absent

UC4 — Compare Error Trends Across Multiple Writing Samples

Primary endpoint: GET /api/students/:studentId/trends

Current automated coverage — Implemented: 4; Partially implemented: 2; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC4-01 prepareTrendSamples	Unsorted server trends with unsure counts	Sorts chronologically; charts five categories; retains all-category total	None	None	None	None
UT-UC4-02 Range and sample filters	Prepared samples; preset/custom range; excluded IDs	Returns only included samples in range	None	None	None	None
UT-UC4-03 summariseTrends	Two or more included samples	Returns totals, most frequent category and improving/steady/more-errors comparison	None	None	None	None
UT-UC4-04 ErrorTrendsPage: insufficient data	Zero or one included sample	Shows insufficient trend guidance	getStudentTrends; getStudent	0/1 trend samples; student	None	None
UT-UC4-05 getTrends: valid student	studentId; optional valid dates	Returns chronological per-sample counts excluding dismissed errors	Student.findById; Sample.find.sort	student; analysed/reviewed Samples	Samples exist	No state change
UT-UC4-06 getTrends: invalid dates	Malformed/impossible/ from-after-to dates	Returns 400 validation error; no Sample query	Student.findById; Sample.find spy	student; unused	No state change	No state change
UT-UC4-07 getTrends: missing student	Unknown studentId	Returns 404 Student not found	Student.findById	null	Student absent	Student absent

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Primary endpoint: PATCH /api/samples/:sampleId/
errors/:errorIndex

Current automated coverage — Implemented: 7; Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC5-01 ErrorCard: reclassify	Existing error; one of five educator categories	Calls save with chosen category	onReclassify	captured callback	Error.category = old	Rendered card shows new category
UT-UC5-02 ErrorCard: remove/restore	Existing live or dismissed error	Confirms dismissal or offers restore	onDismiss/onRestore	captured callback	dismissed = false/true	Rendered state toggles
UT-UC5-03 ErrorCard: confirm uncertainty	Below-threshold confidence	Calls confirmation without changing category	onConfirm	captured callback	confidence < threshold	Rendered error is confirmed
UT-UC5-04 SampleReportPage: add missed error	written text; selected category	Calls updateSampleError(sampleId, new, payload)	updateSampleError	updated Sample	Form ready	Form submitted
UT-UC5-05 reclassifyError: category patch	sampleId; numeric index; valid category	Loads Sample first; changes category; saves once; returns full Sample	Sample.findById; sample.save	Sample; saved Sample	category = old	category = new
UT-UC5-06 reclassifyError: dismiss patch	sampleId; index; dismissed boolean	Preserves category; saves once; returns full Sample	Sample.findById; sample.save	Sample; saved Sample	dismissed = false	dismissed = true
UT-UC5-07 reclassifyError: confidence patch	sampleId; index; confidenceScore = 1	Preserves category/dismissal; saves confirmed score	Sample.findById; sample.save	Sample; saved Sample	confidence < threshold	confidenceScore = 1
UT-UC5-08 reclassifyError: append manual error	errorIndex = new; exact written; valid category	Appends active confidence=1 error; recomputes statistics	Sample.findById; sample.save	Sample; saved Sample	errors = prior list	errors += exact educator error
UT-UC5-09 reclassifyError: invalid request	Missing Sample; bad index/category/type/range/text	Returns 400/404 and does not save	Sample.findById; sample.save spy	null or Sample; unused save	Sample unchanged	Sample unchanged

UC6 — Regenerate Intervention Recommendations

Primary endpoint: POST /api/students/:studentId/recommendations

Current automated coverage — Implemented: 6; Not implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC6-01 Recommendation UI: generate/refresh	studentId; button click	Calls generateRecommendations once; replaces displayed report	generateRecommendations	201 current report	Old/no report displayed	New report displayed
UT-UC6-02 generateReport: active evidence	Student; analysed/reviewed Samples	Excludes dismissed errors; requests up to four strategies; returns sources	RecommendationEngine. createInterventionStrategies	Grounded strategies	No report mutation	No report mutation
UT-UC6-03 generateReport: no active errors	Samples absent or every error dismissed	Throws 422 NO_ACTIVE_ERRORS; engine not called	RecommendationEngine spy	unused	Existing report remains	Existing report remains
UT-UC6-04 RecommendationEngine: valid strategy	Evidence and approved worksheets	Returns validated grounded strategy list	Azure/Gemini adapters	Approved context; structured model output	None	None
UT-UC6-05 RecommendationEngine: external failure	Timeout/transient/ malformed model response	Retries transient failure; returns controlled error for malformed output	Azure/Gemini adapters	Failure sequence	None	None
UT-UC6-06 generateStudent Recommendations: success	Valid studentId with active evidence	Generates first, then performs one atomic latest-only upsert; returns 201	Student.findById; Sample.find; generateReport; RecommendationReport. findOneAndUpdate	student; samples; generated report; current report	Old/no report	One current report
UT-UC6-07 generateStudent Recommendations: missing student	Unknown studentId	Returns 404 STUDENT_NOT_FOUND; no generation or upsert	Student.findById	null	No report change	No report change
UT-UC6-08 Concurrent first-upsert race	findOneAndUpdate throws E11000	Reads and returns winning current report	RecommendationReport. findOneAndUpdate; findOne	E11000; winner	Concurrent creation	One current report
UT-UC6-09 Generation failure preserves old report	Existing report; engine/generateReport rejects	Returns error before persistence; old report unchanged	generateReport; findOneAndUpdate spy	controlled rejection; unused upsert	Old report exists	Old report unchanged

UC7 — Log In

Primary endpoint: POST /api/auth/login

Current automated coverage — Implemented: 3; Partially implemented: 1; Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC7-01 LoginPage: successful submit	username; password	Saves safe account and navigates to requested route/caseload	login; saveSession	safe account; resolved	No client session	Account stored in localStorage
UT-UC7-02 LoginPage: rejected credentials	Unknown user or wrong password	Shows same inline error; does not navigate/save	login; saveSession spy	401 error; unused save	No client session	No client session
UT-UC7-03 authController.login: success	Known username; correct password	Returns safe account only; no JWT/reset/hash fields	Account.findOne; bcrypt.compare	account; true	Account exists	Account unchanged
UT-UC7-04 authController.login: missing input	Missing username or password	Returns 400 validation error	Account.findOne spy	unused	Account unchanged	Account unchanged
UT-UC7-05 authController.login: bad credentials	Unknown username or wrong password	Returns identical 401 Incorrect username or password	Account.findOne; bcrypt.compare	null/Account; false	Account unchanged	Account unchanged
UT-UC7-06 session route gate	safe account or no local session	saveSession persists object; RequireAuth permits or redirects	localStorage	stored account or null	Session absent/present	Session present/unchanged

UC8 — Change Password

Primary endpoint: PATCH /api/auth/change-password

Current automated coverage — Implemented: 1; Not implemented: 5.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC8-01 ChangePasswordForm: valid submit	current; strong new; matching confirmation	Calls changePassword; clears fields; shows Password updated	changePassword	200 success	Fields filled	Fields cleared
UT-UC8-02 ChangePasswordForm: invalid client input	Weak new password or mismatch	Disables submit or shows mismatch; no API call	changePassword spy	unused	Account unchanged	Account unchanged
UT-UC8-03 passwordPolicy: strength matrix	Passwords at/around each rule boundary	Returns all unmet rules; accepts exact valid boundary	None	None	None	None
UT-UC8-04 changePassword: valid	Known username; correct current; strong new	Hashes new password; saves once; returns success	Account.findOne; bcrypt.compare/hash; account.save	Account; true; new hash; saved	Old passwordHash	New passwordHash
UT-UC8-05 changePassword: wrong current	Unknown account or wrong current password	Returns same 401; does not hash/save	Account.findOne; bcrypt.compare; hash/save spies	null/Account; false; unused	Old passwordHash	Old passwordHash
UT-UC8-06 changePassword: weak new	Correct current; weak new password	Returns 400 requirements; does not hash/save	Account.findOne; bcrypt.compare; hash/save spies	Account; true; unused	Old passwordHash	Old passwordHash

UC9 — Reset a Forgotten Password

Primary endpoint: POST /api/auth/forgot-password + POST /api/auth/reset-password/
:token

Current automated coverage — Implemented: 5; Not implemented: 4.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC9-01 ForgotPasswordPage: accepted	username	Shows Check your email	requestPasswordReset	200 generic response	phase = form	phase = sent
UT-UC9-02 ForgotPasswordPage: request error	username; provider/server rejection	Shows inline error; remains form	requestPasswordReset	request error	phase = form	phase = form
UT-UC9-03 forgotPassword: known account	Known username	Creates 64-hex token with 1-hour expiry; saves; sends link; returns generic 200	Account.findOne; account.save; sendPasswordResetEmail	Account; saved; email accepted	No reset token	Valid token + expiry
UT-UC9-04 forgotPassword: unknown account	Unknown username	Returns identical generic 200; sends no email	Account.findOne; send email spy	null; unused	No account	No account
UT-UC9-05 ResetPasswordPage: client validation	Weak or mismatched password	Disables submit or shows guidance; API not called	resetPassword spy	unused	phase = form	phase = form
UT-UC9-06 resetPassword: valid token	Unexpired token; strong password	Hashes password; clears token fields; saves; returns 200	Account.findOne; bcrypt.hash; account.save	Account; hash; saved	Token valid; old hash	Token null; expiry null; new hash
UT-UC9-07 resetPassword: weak password	Valid token; weak password	Returns 400 before token lookup/consumption	Account.findOne spy	unused	Token valid	Token remains valid
UT-UC9-08 resetPassword: invalid/expired token	Unknown or expired token; strong password	Returns 400 invalid/expired; no hash/save	Account.findOne; bcrypt/hash spies	null; unused	No valid token	No state change
UT-UC9-09 emailService: Resend adapter	Recipient and resetLink	Posts expected email; throws controlled provider failure	fetch(Resend API)	2xx JSON or non-2xx body	None	None

3. Bottom-Up Integration Test Tables

Core/persistence integration precedes route-level integration. External AI, storage, email and synthetic filesystem boundaries remain mocked.

UC1 — Upload Student Writing Sample for Analysis

Integration Plan A — Analysis Engine + Sample Persistence

Real Components: errorClassificationEngine, Sample model, isolated MongoDB

Mock Components: Gemini API and synthetic filesystem only

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC1-01 Persist successful analysis	UPLOADED synthetic Sample	Real runAnalysis saves ANALYSED and intact errors	Gemini generateContent	Valid JSON	Sample = UPLOADED	Sample = ANALYSED with errors
BI-UC1-02 Persist failed analysis	Unreadable synthetic page	Real runAnalysis saves FAILED and safe reason	Filesystem read/Gemini	Read/model failure	Sample = UPLOADED	Sample = FAILED with analysisError

Integration Plan B — Upload Route + Controller + Core

Real Components: Express samples route, Multer, controller, models, isolated MongoDB, analysis service

Mock Components: Gemini API only

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC1-03 Accepted upload then background completion	Valid student + synthetic image	POST returns 202 first; later GET reaches ANALYSED	Gemini generateContent	Valid JSON	Student exists; no Sample	Sample created then ANALYSED
BI-UC1-04 Rejected upload is atomic	Unknown student or invalid second file	400/422; no Sample; all temporary files removed	Filesystem content/cleanup	Invalid file + successful cleanup	No Sample	No Sample or orphan files

UC2 — View Categorised Error Pattern Report

Integration Plan A — Sample Model + Serializer

Real Components: Sample model, toClientSample, isolated MongoDB

Mock Components: None

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC2-01 Serialize report and counts	Persisted Sample with live/dismissed/unsure errors	Public shape has exact counts and no imagePath	None	None	Sample exists	No state change
BI-UC2-02 Resolve private scan internally	Sample page and valid/invalid index	Valid bytes served; invalid index 404; path not serialized	Synthetic filesystem	Synthetic image bytes	Sample exists	No state change

Integration Plan B — GET Sample/Image HTTP Stack

Real Components: Express route, controller, Sample model, isolated MongoDB

Mock Components: Synthetic filesystem only

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC2-03 Read existing report	GET sample and image endpoints	200 report then image bytes; confidence/state retained	Synthetic filesystem	Image bytes	Sample exists	No state change
BI-UC2-04 Read missing report/image	Unknown sample or invalid page index	404 through error middleware; no private path	None	None	Sample absent or page missing	No state change

UC3 — View Recommended Intervention Strategies with Rationale

Integration Plan A — Report Freshness Read

Real Components: RecommendationReport and Sample models, isolated MongoDB

Mock Components: None

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC3-01 Current report read	Report newer than all evidence	Returns isOutdated false; no write	None	None	One report exists	Same report unchanged
BI-UC3-02 Changed evidence read	Sample updated or added after generatedAt	Returns same report with isOutdated true; no write	None	None	Report + changed Sample	Records unchanged

Integration Plan B — GET Latest HTTP Stack

Real Components: Express route, controller, models, isolated MongoDB

Mock Components: None

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC3-03 Latest report success	GET latest for current/stale report	200 report with computed flag; generation never called	Recommendation engine spy	Unused	Report exists	No state change
BI-UC3-04 No report or malformed ID	Unknown valid ID or malformed ID	404 RECOMMENDATION_NOT_FOUND or 400 INVALID_ID	None	None	Report absent	Report absent

UC4 — Compare Error Trends Across Multiple Writing Samples

Integration Plan A — Trend Query + Aggregation

Real Components: Student and Sample models, studentController, isolated MongoDB

Mock Components: None

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC4-01 Aggregate analysed/reviewed samples	Mixed statuses and dismissed/unsure errors	Chronological trends exclude dismissed; totals include unsure	None	None	Synthetic Samples exist	No state change
BI-UC4-02 Inclusive date bounds	Samples on from/to day edges	Includes full requested days in UTC	None	None	Boundary Samples exist	No state change

Integration Plan B — Trends HTTP Stack

Real Components: Express student route, controller, models, isolated MongoDB

Mock Components: None

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC4-03 GET all-time trends	Valid studentId	200 totals and per-sample trends consumed by client contract	None	None	Student + Samples exist	No state change
BI-UC4-04 Reject bad request	Missing student or invalid dates	404 or 400; no partial response	None	None	No mutation	No mutation

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Integration Plan A — Embedded Error Persistence

Real Components: Sample model, reclassifyError, isolated MongoDB

Mock Components: None

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC5-01 Patch existing embedded error	Valid category/dismissed/ confidence patch	Loads then mutates one embedded item; preserves other fields	None	None	Sample has original error	Sample has patched error
BI-UC5-02 Append educator error	errorIndex=new; exact written; valid category	Appends active confidence=1 error; statistics increase	None	None	Sample.errors = prior list	Sample.errors += exact text

Integration Plan B — PATCH Error HTTP Stack

Real Components: Express samples route, controller, Sample model, isolated MongoDB

Mock Components: None

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC5-03 All supported patch forms	Four PATCH requests: category, dismissed, confidence, new	Each returns authoritative full Sample and expected state	None	None	Sample exists	Four expected persisted updates
BI-UC5-04 Invalid patch is non-mutating	Bad index/category/type/ range/text or missing Sample	400/404; database unchanged	None	None	Baseline Sample	Baseline Sample unchanged

UC6 — Regenerate Intervention Recommendations

Integration Plan A — Generation + Atomic Report Store

Real Components: generateReport, RecommendationEngine, models, isolated MongoDB

Mock Components: Gemini and Azure only

Current automated coverage — Partially implemented: 2; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC6-01 Successful atomic replacement	Existing report; active analysed evidence	Generates then one atomic upsert leaves one latest report	Gemini/Azure	Valid grounded strategy	Old report exists	One replaced current report
BI-UC6-02 Generation failure retains old report	Existing report; external/generation failure	No upsert; previous report byte-for-byte unchanged	Gemini/Azure	Controlled failure	Old report exists	Old report unchanged
BI-UC6-03 Concurrent first upsert	Two first-generation requests	Duplicate-key loser reads and returns winner; one report remains	Gemini/Azure	Valid strategy	No report	Exactly one current report

Integration Plan B — POST Recommendation HTTP Stack

Real Components: Express route, controller, services, models, isolated MongoDB

Mock Components: Gemini and Azure only

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC6-04 Generate current report	Valid student with active errors	201 report; subsequent GET latest returns same report	Gemini/Azure	Valid strategy	No/old report	Current report stored
BI-UC6-05 Reject missing/no-evidence student	Missing student or no active errors	404/422; existing report unchanged	Gemini/Azure spy	Unused	Old/no report	Unchanged

UC7 — Log In

Integration Plan A — Account Lookup + Password Check

Real Components: authController, Account model, bcrypt, isolated MongoDB

Mock Components: None

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC7-01 Valid credentials	Known account and correct password	Safe account response; no hash/reset/token fields	None	None	Account exists	Account unchanged
BI-UC7-02 Invalid credentials	Unknown user or wrong password	Same 401 response for both; no state change	None	None	Account absent/present	No state change

Integration Plan B — Login HTTP Stack

Real Components: Express auth route, controller, Account model, isolated MongoDB

Mock Components: None

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC7-03 POST login success	Valid JSON credentials	200 safe account object; no JWT issued	None	None	No client session	Server account unchanged
BI-UC7-04 POST login validation	Missing field, unknown user, wrong password	400 or generic 401 through middleware	None	None	Account unchanged	Account unchanged

UC8 — Change Password

Integration Plan A — Password Policy + Account Persistence

Real Components: authController, passwordPolicy, Account model, bcrypt, isolated MongoDB

Mock Components: None

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC8-01 Change valid password	Correct current; strong new password	New hash persists and old password no longer matches	None	None	Old passwordHash	New passwordHash
BI-UC8-02 Reject incorrect/weak change	Wrong current or weak new password	401/400; hash unchanged	None	None	Old passwordHash	Old passwordHash

Integration Plan B — Change-Password HTTP Stack

Real Components: Express auth route, controller, Account model, isolated MongoDB

Mock Components: None

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC8-03 PATCH success	Valid username/current/new body	200; subsequent login accepts new and rejects old password	None	None	Old credential valid	New credential valid
BI-UC8-04 PATCH failure paths	Missing/unknown/wrong/weak values	400/401; generic current-password error; no mutation	None	None	Account unchanged	Account unchanged

UC9 — Reset a Forgotten Password

Integration Plan A — Reset Token + Account Persistence

Real Components: authController, Account model, bcrypt, isolated MongoDB

Mock Components: Email service only

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC9-01 Issue reset token	Known username	Stores token/expiry then requests email; generic 200	sendPasswordResetEmail	Accepted	No reset token	Valid token + 1-hour expiry
BI-UC9-02 Consume valid token	Valid token; strong password	Updates hash; clears token/expiry atomically on save	None	None	Valid token; old hash	No token; new hash

Integration Plan B — Forgot/Reset HTTP Stack

Real Components: Express auth routes, controllers, Account model, isolated MongoDB

Mock Components: Resend API only

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC9-03 Forgot-password privacy pair	Existing and missing username requests	Both return identical 200; email only for existing account	Resend API	Accepted	Known account has no token	Known account gets token
BI-UC9-04 Reset success and rejection	Valid, weak, invalid and expired token requests	Correct 200/400 responses; rejected requests preserve state	None	None	Prepared token states	Only valid token consumed

4. System End-to-End Test Tables

Real browser + client + Express + isolated MongoDB. External services use deterministic doubles. No automated browser E2E suite was found.

UC1 — Upload Student Writing Sample for Analysis

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC1-01 Upload and analyse successfully	Signed-in teacher; student exists; synthetic valid page; deterministic AI	Open student → Upload sample → choose file/task → Analyse → wait	202 accepted; analysing screen; success state; Open error report enabled	No Sample	Sample = ANALYSED with errors	No automated browser E2E test found.
E2E-UC1-02 Reject invalid upload	Signed-in teacher; student exists; invalid synthetic file	Submit invalid upload	Inline 400/422 error; entered form retained; no Sample	No Sample	No Sample	No automated browser E2E test found.
E2E-UC1-03 Analysis failure and polling timeout states	Synthetic analysis failure or non-terminal polling fixture	Upload; observe failure. Repeat with stalled polling; choose Check again	FAILED shows safe reason; timeout offers Check again/return	No Sample	FAILED or still-UPLOADED test Sample	No automated browser E2E test found.

UC2 — View Categorised Error Pattern Report

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC2-01 View categorised report	ANALYSED synthetic Sample with live/dismissed/uncertain/multi-page errors	Open report; filter category; switch page; select box; reveal removed tags	Correct groups/counts/uncertainty/scan boxes; no raw imagePath	Sample exists	No persistent change	No automated browser E2E test found.
E2E-UC2-02 Missing report or image	Unknown sample or invalid synthetic image index	Open report/image URL	404 state or image fallback; no private path disclosed	Sample absent/invalid page	No state change	No automated browser E2E test found.

UC3 — View Recommended Intervention Strategies with Rationale

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC3-01 View current recommendations	Current RecommendationReport exists	Open Recommendations	Existing strategies/evidence/ rationale displayed; no POST generation	Current report exists	Same report unchanged	No automated browser E2E test found.
E2E-UC3-02 View no-report state	Analysed sample exists but report absent	Open Recommendations	Generate recommendations action displayed; no automatic POST	No report	No report	No automated browser E2E test found.
E2E-UC3-03 View outdated report	Report exists; source Sample updated later	Open Recommendations	Old report displayed with Refresh now warning; no automatic regeneration	Old report + changed Sample	Same report unchanged	No automated browser E2E test found.

UC4 — Compare Error Trends Across Multiple Writing Samples

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC4-01 Compare multiple samples	At least three analysed/reviewed samples with synthetic errors	Open trends; choose range; exclude/reinclude sample	Chart and summary update locally; totals reconcile with report counts	Samples exist	No persistent change	No automated browser E2E test found.
E2E-UC4-02 Insufficient comparison	Zero/one sample remains after filtering	Open trends; narrow range or exclude samples	Insufficient trend guidance; no misleading comparison	Samples exist	No persistent change	No automated browser E2E test found.
E2E-UC4-03 Missing student	Unknown student route	Open trends URL	Student not found shown	Student absent	Student absent	No automated browser E2E test found.

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC5-01 Correct existing AI tags	Report has reclassifiable, dismissible and uncertain errors	Reclassify one; remove one; confirm one	Each card/count updates from full server Sample; written text unchanged	Original embedded errors	Category/dismissed/confidence updated	No automated browser E2E test found.
E2E-UC5-02 Restore removed tag	Report has dismissed error	Show removed tags → Restore	Error returns to live group and counts	dismissed = true	dismissed = false	No automated browser E2E test found.
E2E-UC5-03 Add an AI-missed error	Report exists	Enter exact child-written text; select one of five categories; Add missed error	New educator error appears with confidence 1; counts increase	No manual error	Exact manual error appended	No automated browser E2E test found.
E2E-UC5-04 Rejected correction	Inject missing Sample or invalid patch response	Attempt correction	Inline failure; UI does not claim success	Sample/UI baseline	Baseline retained	No automated browser E2E test found.

UC6 — Regenerate Intervention Recommendations

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC6-01 Generate first recommendations	Student has active reviewed errors; no report	Open Recommendations → Generate	201 report appears with up to four grounded strategies	No report	One current report	No automated browser E2E test found.
E2E-UC6-02 Refresh outdated recommendations	Old report and newer corrected Sample	Open Recommendations → Refresh now	Displayed report replaced; subsequent GET is not outdated	Old report	One replaced current report	No automated browser E2E test found.
E2E-UC6-03 No active errors	All errors dismissed or no analysed errors	Select Generate/Refresh	Visible NO_ACTIVE_ERRORS failure; old report retained if present	Old/no report	Unchanged	No automated browser E2E test found.
E2E-UC6-04 External generation failure	Existing report; deterministic upstream failure	Select Refresh	Visible controlled failure; previous report remains displayed/stored	Old report	Old report unchanged	No automated browser E2E test found.

UC7 — Log In

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC7-01 Log in successfully	Known synthetic account; no session	Open login; submit valid credentials	Navigates to requested route/caseload; safe account in localStorage	No session	Client session present	No automated browser E2E test found.
E2E-UC7-02 Reject invalid login	No session	Submit unknown username; repeat with wrong password	Same inline error; remains logged out	No session	No session	No automated browser E2E test found.
E2E-UC7-03 Protected route gate	No session then valid local session	Open protected route; then log in and retry	Redirects to login when absent; allows route when present	Session absent	Session present after login	No automated browser E2E test found.

UC8 — Change Password

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC8-01 Change password successfully	Signed-in synthetic account	Open Account; enter current/strong matching new password; submit; sign out/in	Password updated shown; old login fails; new login succeeds	Old hash	New hash	No automated browser E2E test found.
E2E-UC8-02 Client rejects weak/mismatch	Signed-in account	Enter weak or mismatched new values	Submit disabled/guidance visible; no PATCH	Old hash	Old hash	No automated browser E2E test found.
E2E-UC8-03 Server rejects wrong current	Signed-in account	Enter wrong current and valid new values; submit	Inline current-password error; remains signed in; hash unchanged	Old hash	Old hash	No automated browser E2E test found.

UC9 — Reset a Forgotten Password

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC9-01 Forgot and reset successfully	Known account; test email capture enabled	Request reset → open captured link → enter strong matching password	Generic sent state then Password updated; new login works	Old hash; no token	New hash; token cleared	No automated browser E2E test found.
E2E-UC9-02 Unknown username privacy	Unknown username	Request password reset	Same generic Check your email state; no message sent	No account	No account	No automated browser E2E test found.
E2E-UC9-03 Reject invalid/expired link	Invalid or expired token	Open reset link; submit strong matching password	Inline invalid/expired error; no account mutation	No valid token	No mutation	No automated browser E2E test found.
E2E-UC9-04 Reject weak/mismatch locally	Valid token	Enter weak or mismatched passwords	Submit disabled/guidance visible; token remains usable	Valid token	Valid token	No automated browser E2E test found.

5. Robustness Test Tables

Cases are limited to sequence branches, explicit validations, external boundaries and state-integrity invariants.

UC1 — Upload Student Writing Sample for Analysis

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC1-01 Upload validation	No files, unsupported bytes, corrupt PDF, unknown student	Multer/toPageFiles/ createSample	400/422 safe message; all temp files cleaned	No Sample or orphan files	Teacher can correct form and retry	Partially implemented Engine unsupported/corrupt tests exist; controller upload atomicity/cleanup tests are missing.
RB-UC1-02 Multi-file atomicity	Second file fails after first validates	toPageFiles Promise.all	Whole request fails; first file is not registered	Samples remain unchanged	Retry with valid set	Not implemented No matching automated test found.
RB-UC1-03 AI output validation	Unknown category, bad confidence/location/page, missing written	validateAnalysisResult	Rejects result; retries; eventually FAILED	No invalid errors persisted	Later re-upload/retry possible	Implemented errorClassificationEngine. test.js covers all listed schema faults.
RB-UC1-04 AI timeout/retry	Hung/transient/ rate-limited Gemini call	callModelWithRetry	Bounded retries/backoff; safe FAILED result after exhaustion	Sample ends terminal, never stuck by normal failure path	Teacher sees failure and can retry	Implemented errorClassificationEngine. test.js retry/rate-limit/timeout tests.
RB-UC1-05 Polling resilience	One dropped GET then recovery; 30 non-terminal attempts	useSamplePolling	Transient failure tolerated; bounded timeout after ceiling	Server analysis continues independently	Check again restarts ceiling	Not implemented No matching automated test found.

UC2 — View Categorised Error Pattern Report

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC2-01 Sample lookup	Unknown sampleId	getSample	404; no partial report	No mutation	Return to profile	Not implemented No matching automated test found.
RB-UC2-02 Image boundary	Negative/out-of-range page index or unreadable image	getImages/ScanViewer	404 or image fallback; no raw path	No mutation	Other pages/report remain usable	Partially implemented ScanViewer image-load fallback is tested; server index/path boundary is not.
RB-UC2-03 Sensitive-path disclosure	Sample contains private imagePath	toClientSample and errors	JSON omits imagePath; image only via endpoint	Private path remains server-side	None	Partially implemented Several controller fixtures include private paths, but no direct getSample disclosure assertion found.
RB-UC2-04 Uncertainty semantics	Below-threshold score with valid category	ErrorCard/report grouping	Uncertain visual state without changing category	Sample unchanged	Teacher may confirm through UC5	Implemented ErrorCard uncertain-state test and engine threshold tests.

UC3 — View Recommended Intervention Strategies with Rationale

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC3-01 Invalid identifier	Malformed studentId	assertObjectId	400 INVALID_ID; model not queried	No mutation	Use valid student route	Implemented recommendationRoutes.test.js validates malformed IDs.
RB-UC3-02 Absent report	Valid studentId with no report	RecommendationReport.findOne	404 mapped by client to no-report state	No report	Teacher may explicitly generate	Partially implemented Client mapping/controller 404 branch has no direct automated test.
RB-UC3-03 Freshness race	Sample updated/added after generatedAt	Sample.exists freshness query	isOutdated true on read; report is not mutated	Old report retained	Teacher chooses Refresh	Implemented recommendationRoutes.test.js covers changed and newly usable evidence query.
RB-UC3-04 Read-only invariant	Current/outdated GET latest	Controller/model spies	No generateReport/engine/upsert call	Report exists	Report unchanged	Not implemented No matching automated test found.

UC4 — Compare Error Trends Across Multiple Writing Samples

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC4-01 Date validation	Malformed, impossible, or from > to	getTrends query parser	400; no misleading trend set	No mutation	Correct dates and retry	Implemented students.integration.test.js date validation tests.
RB-UC4-02 Boundary dates	Samples late on requested to date	UTC date query	Entire to day included	No mutation	None	Implemented students.integration.test.js — includes entire requested to date.
RB-UC4-03 Count reconciliation	Dismissed and unsure errors mixed with five categories	Server aggregation + prepareTrendSamples	Dismissed excluded; unsure omitted from lines but included total	No mutation	None	Partially implemented client trends reconciliation is tested; server dismissed aggregation is not directly covered.
RB-UC4-04 Sparse selection	Zero/one sample after local filters	summariseTrends/ErrorTrendsPage	No fabricated comparison; guidance shown	No mutation	Reinclude samples/change range	Partially implemented Pure summary handles data; page guidance lacks automated test.

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC5-01 Index validation	Negative, non-integer, or out-of-range errorIndex	reclassifyError	404 Error not found; no save	Sample unchanged	Refresh report	Partially implemented sampleController.test.js covers out-of-range positive index only.
RB-UC5-02 Patch-field validation	No supported field; invalid category; non-boolean dismissed; score outside 0..1	reclassifyError	400; no partial mutation/save	Sample unchanged	Correct field and retry	Not implemented No matching automated test found.
RB-UC5-03 Manual text validation	Empty/whitespace written or invalid category	errorIndex=new branch	400; no appended error	Errors unchanged	Enter exact text/category and retry	Not implemented No matching automated test found.
RB-UC5-04 Child-text integrity	Written text includes misspelling/spacing/case	Manual append and subsequent serialization	Preserves submitted written string exactly	No manual error	Exact string stored	Partially implemented sampleController.new-error test asserts a misspelling; broader whitespace/case integrity is not tested.
RB-UC5-05 Save failure	Database save rejects after in-memory patch	sample.save	Error response; UI shows failure and must not claim success	Persisted Sample unchanged	Reload authoritative Sample/retry	Not implemented No matching automated test found.

UC6 — Regenerate Intervention Recommendations

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC6-01 Missing/no-evidence input	Unknown student or no active errors	Controller/generateReport	404 or 422; no upsert	Old/no report	Add/review evidence then retry	Partially implemented generateReport no-active-errors test exists; missing-student generation test missing.
RB-UC6-02 External service failure	Azure timeout/oversize or Gemini transient/malformed output	RecommendationEngine adapters	Controlled failure; no report persistence	Old report exists	Old report remains; retry later	Partially implemented Engine faults are tested; old-report persistence invariant is not.
RB-UC6-03 Grounding integrity	Unknown evidence/worksheet IDs or unsupported category	RecommendationEngine validation	Rejects model output; no upsert	Old/no report	Fix external output/retry	Implemented recommendationEngine.test.js evidence, worksheet and category validation tests.
RB-UC6-04 Atomic replacement	Failure between generation and persistence	Controller persistence boundary	No half-superseded state; latest report stays old or becomes wholly new	One old report	Exactly one valid latest report	Partially implemented Atomic findOneAndUpdate options tested; real failure atomicity not integrated.
RB-UC6-05 First-upsert race	Concurrent POST requests on absent report	findOneAndUpdate duplicate-key branch	One winner; loser returns winner; no duplicate reports	No report	Exactly one report	Partially implemented recommendationRoutes.test.js simulates E11000 recovery.

UC7 — Log In

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC7-01 Credential validation	Missing username/password	login controller	400 validation error; no lookup	No session/account mutation	Complete fields and retry	Implemented auth.integration.test.js missing-password case.
RB-UC7-02 Enumeration resistance	Unknown username versus wrong password	login controller	Same 401 message for both	No mutation	Retry valid credentials	Implemented auth.integration.test.js asserts same message in separate cases.
RB-UC7-03 Response secrecy	Valid login account includes hash/reset fields internally	toClientAccount	Only safe account fields returned; no JWT invented	Account unchanged	Client stores safe object only	Partially implemented Successful integration test asserts exact username/email response; no broader client session payload test.
RB-UC7-04 Client-only gate limitation	Direct API call without local session	Server API boundary	Request remains public per sequence; test must not expect JWT enforcement	No session	No state change	Not implemented No matching automated test found.

UC8 — Change Password

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC8-01 Current-password secrecy	Unknown account versus wrong current password	changePassword controller	Same 401 message; no hash/save	Old hash or no account	Unchanged	Not implemented No matching automated test found.
RB-UC8-02 Server-side strength enforcement	Client bypass with weak new password	passwordPolicy in controller	400 requirements; no hash/save	Old hash	Old hash	Partially implemented Password policy is unit-tested; controller integration is missing.
RB-UC8-03 Confirmation mismatch	Strong new and different confirmation	ChangePasswordForm	Submit disabled/guidance; no API call	Old hash	Old hash	Not implemented No matching automated test found.
RB-UC8-04 Persistence failure	bcrypt succeeds but account.save rejects	account.save	Error response; old persisted hash remains	Old hash	Old hash	Not implemented No matching automated test found.
RB-UC8-05 Boundary password	Exactly 8 chars with required classes	client/server policy	Accepted consistently by both layers	Old hash	New hash after valid change	Partially implemented Server policy boundary is tested; client form consistency and controller path are not.

UC9 — Reset a Forgotten Password

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC9-01 Username privacy	Existing versus unknown username	forgotPassword	Identical generic 200; email only for existing account	No reset token	Only known account gains token	Implemented auth.integration.test.js privacy-pair tests.
RB-UC9-02 Email provider failure	Resend non-2xx/network failure	emailService after token save	Controlled server error; no false sent state	Token may already be stored per sequence order	Teacher can retry; old token expires	Not implemented No matching automated test found.
RB-UC9-03 Weak password	Valid token with weak password	resetPassword policy before lookup	400; token not consumed	Valid token	Valid token	Implemented auth.integration.test.js weak password does not consume token.
RB-UC9-04 Invalid/expired/reused token	Unknown, expired, or previously consumed token	Account.findOne token query	400 invalid or expired; no hash/save	No valid token	No state change	Partially implemented Unknown/expired is tested; explicit token reuse after success is not.
RB-UC9-05 Successful token consumption	Valid token and strong password	bcrypt + account.save	New hash saved; token and expiry cleared	Valid token; old hash	No token; new hash	Implemented auth.integration.test.js success case.

6. Sequence-to-Test Traceability and Coverage

Every case maps to a corrected sequence branch. Existing evidence names the closest automated test.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC1-01	UC1	Unit	Valid upload	Not implemented	No matching automated test found.
UT-UC1-02	UC1	Unit	Student missing or upload invalid	Not implemented	No matching automated test found.
UT-UC1-03	UC1	Unit	Polling outcomes	Not implemented	No matching automated test found.
UT-UC1-04	UC1	Unit	Valid upload	Not implemented	No matching automated test found.
UT-UC1-05	UC1	Unit	Student missing or upload invalid	Not implemented	No matching automated test found.
UT-UC1-06	UC1	Unit	Analysis succeeds	Implemented	server/services/ errorClassificationEngine. test.js — valid result, category, confidence, prompt and preservation tests.
UT-UC1-07	UC1	Unit	Gemini analysis mode	Implemented	server/services/ errorClassificationEngine. test.js — retry, malformed output, page-index, timeout and maxRetries tests.
UT-UC1-08	UC1	Unit	Analysis succeeds / fails	Implemented	server/services/ errorClassificationEngine. integration.test.js — IT-01, IT-02, IT-03 and IT-04.
E2E-UC1-01	UC1	System E2E	Upload and analyse successfully	Not implemented	No automated browser E2E test found.
E2E-UC1-02	UC1	System E2E	Reject invalid upload	Not implemented	No automated browser E2E test found.
E2E-UC1-03	UC1	System E2E	Analysis failure and polling timeout states	Not implemented	No automated browser E2E test found.
RB-UC1-01	UC1	Robustness	Upload validation	Partially implemented	Engine unsupported/corrupt tests exist; controller upload atomicity/cleanup tests are missing.
RB-UC1-02	UC1	Robustness	Multi-file atomicity	Not implemented	No matching automated test found.
RB-UC1-03	UC1	Robustness	AI output validation	Implemented	errorClassificationEngine. test.js covers all listed schema faults.
RB-UC1-04	UC1	Robustness	AI timeout/retry	Implemented	errorClassificationEngine. test.js retry/rate-limit/timeout tests.
RB-UC1-05	UC1	Robustness	Polling resilience	Not implemented	No matching automated test found.
BI-UC1-01	UC1	Bottom-up integration	Analysis succeeds	Partially implemented	errorClassificationEngine. integration.test.js exercises persistence with patched model behavior; not an isolated real MongoDB.
BI-UC1-02	UC1	Bottom-up integration	Analysis fails	Partially implemented	errorClassificationEngine. integration.test.js IT-02; persistence is stub-oriented.
BI-UC1-03	UC1	Bottom-up integration	Valid upload and polling	Not implemented	No matching automated test found.
BI-UC1-04	UC1	Bottom-up integration	Student missing or upload invalid	Not implemented	No matching automated test found.
UT-UC2-01	UC2	Unit	Sample exists	Implemented	client/test/ SampleReportPage.test.js — grouping, dismissed-error and filter tests.
UT-UC2-02	UC2	Unit	confidenceScore < confidenceThreshold	Implemented	client/test/ErrorCard. test.js — uncertain state flags below-threshold confidence and confirms via onConfirm.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC2-03	UC2	Unit	For each displayed scan page	Implemented	client/test/ScanViewer. test.js — box filtering, click, thumbnails, zoom, overlay and image-failure tests.
UT-UC2-04	UC2	Unit	Sample exists	Not implemented	No matching automated test found.
UT-UC2-05	UC2	Unit	Sample not found	Not implemented	No matching automated test found.
UT-UC2-06	UC2	Unit	Display scan page	Not implemented	No matching automated test found.
UT-UC2-07	UC2	Unit	compute Statistics(non-dismissed errors)	Not implemented	No matching automated test found.
E2E-UC2-01	UC2	System E2E	View categorised report	Not implemented	No automated browser E2E test found.
E2E-UC2-02	UC2	System E2E	Missing report or image	Not implemented	No automated browser E2E test found.
RB-UC2-01	UC2	Robustness	Sample lookup	Not implemented	No matching automated test found.
RB-UC2-02	UC2	Robustness	Image boundary	Partially implemented	ScanViewer image-load fallback is tested; server index/path boundary is not.
RB-UC2-03	UC2	Robustness	Sensitive-path disclosure	Partially implemented	Several controller fixtures include private paths, but no direct getSample disclosure assertion found.
RB-UC2-04	UC2	Robustness	Uncertainty semantics	Implemented	ErrorCard uncertain-state test and engine threshold tests.
BI-UC2-01	UC2	Bottom-up integration	Sample exists	Not implemented	No matching automated test found.
BI-UC2-02	UC2	Bottom-up integration	Display scan page	Not implemented	No matching automated test found.
BI-UC2-03	UC2	Bottom-up integration	Sample exists	Not implemented	No matching automated test found.
BI-UC2-04	UC2	Bottom-up integration	Sample not found	Not implemented	No matching automated test found.
UT-UC3-01	UC3	Unit	Current report exists	Not implemented	No matching automated test found.
UT-UC3-02	UC3	Unit	No recommendation report exists	Not implemented	No matching automated test found.
UT-UC3-03	UC3	Unit	isOutdated = true	Not implemented	No matching automated test found.
UT-UC3-04	UC3	Unit	Current report exists	Partially implemented	server/test/ recommendationRoutes. test.js tests stale=true but not current=false or explicit no-write behavior.
UT-UC3-05	UC3	Unit	isOutdated = true	Implemented	server/test/ recommendationRoutes. test.js — latest report is marked outdated when usable sample evidence changed.
UT-UC3-06	UC3	Unit	No recommendation report exists	Partially implemented	server/test/ recommendationRoutes. test.js covers malformed ID; no exact latest-report 404 test found.
E2E-UC3-01	UC3	System E2E	View current recommendations	Not implemented	No automated browser E2E test found.
E2E-UC3-02	UC3	System E2E	View no-report state	Not implemented	No automated browser E2E test found.
E2E-UC3-03	UC3	System E2E	View outdated report	Not implemented	No automated browser E2E test found.
RB-UC3-01	UC3	Robustness	Invalid identifier	Implemented	recommendationRoutes. test.js validates malformed IDs.
RB-UC3-02	UC3	Robustness	Absent report	Partially implemented	Client mapping/controller 404 branch has no direct automated test.
RB-UC3-03	UC3	Robustness	Freshness race	Implemented	recommendationRoutes. test.js covers changed and newly usable evidence query.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
RB-UC3-04	UC3	Robustness	Read-only invariant	Not implemented	No matching automated test found.
BI-UC3-01	UC3	Bottom-up integration	Current report exists	Not implemented	No matching automated test found.
BI-UC3-02	UC3	Bottom-up integration	isOutdated = true	Partially implemented	recommendationRoutes. test.js covers query calculation with mocked models, not real database integration.
BI-UC3-03	UC3	Bottom-up integration	Current report exists	Not implemented	No matching automated test found.
BI-UC3-04	UC3	Bottom-up integration	No report exists	Partially implemented	recommendationRoutes. test.js covers malformed ID only.
UT-UC4-01	UC4	Unit	Prepare trend samples	Implemented	client/test/trends.test. js — chronological order and five-category reconciliation tests.
UT-UC4-02	UC4	Unit	Choose date range and subset	Partially implemented	client/test/trends.test. js covers preset range and explicit exclusion; custom-range helper has no direct test.
UT-UC4-03	UC4	Unit	At least 2 included samples	Implemented	client/test/trends.test. js — tie order and all three comparison states.
UT-UC4-04	UC4	Unit	Fewer than 2 included samples	Not implemented	No matching automated test found.
UT-UC4-05	UC4	Unit	Student exists	Partially implemented	server/test/students. integration.test.js covers inclusive to date; no full aggregation/dismissed test found.
UT-UC4-06	UC4	Unit	Validate optional from/to	Implemented	server/test/students. integration.test.js — malformed, impossible and reversed date tests.
UT-UC4-07	UC4	Unit	Student missing	Implemented	server/test/students. integration.test.js — GET trends reports an unknown student as 404.
E2E-UC4-01	UC4	System E2E	Compare multiple samples	Not implemented	No automated browser E2E test found.
E2E-UC4-02	UC4	System E2E	Insufficient comparison	Not implemented	No automated browser E2E test found.
E2E-UC4-03	UC4	System E2E	Missing student	Not implemented	No automated browser E2E test found.
RB-UC4-01	UC4	Robustness	Date validation	Implemented	students.integration. test.js date validation tests.
RB-UC4-02	UC4	Robustness	Boundary dates	Implemented	students.integration. test.js — includes entire requested to date.
RB-UC4-03	UC4	Robustness	Count reconciliation	Partially implemented	client trends reconciliation is tested; server dismissed aggregation is not directly covered.
RB-UC4-04	UC4	Robustness	Sparse selection	Partially implemented	Pure summary handles data; page guidance lacks automated test.
BI-UC4-01	UC4	Bottom-up integration	Student exists	Not implemented	No matching automated test found.
BI-UC4-02	UC4	Bottom-up integration	Validate optional dates	Partially implemented	students.integration. test.js covers inclusive to date with model stubs.
BI-UC4-03	UC4	Bottom-up integration	Student exists	Not implemented	No matching automated test found.
BI-UC4-04	UC4	Bottom-up integration	Student missing / date invalid	Partially implemented	students.integration. test.js covers missing student and date validation over real HTTP with stubbed persistence.
UT-UC5-01	UC5	Unit	Reclassify existing error	Implemented	client/test/ErrorCard. test.js — reclassify panel saves a different category.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC5-02	UC5	Unit	Remove or restore existing error	Implemented	client/test/ErrorCard.test.js — remove confirmation and removed-state restore tests.
UT-UC5-03	UC5	Unit	Confirm uncertain error	Implemented	client/test/ErrorCard.test.js — uncertain state confirmation.
UT-UC5-04	UC5	Unit	Record an error missed by AI	Implemented	client/test/SampleReportPage.test.js — educator can submit an error the AI missed.
UT-UC5-05	UC5	Unit	Valid existing-error patch	Not implemented	No matching automated test found.
UT-UC5-06	UC5	Unit	Remove existing error	Implemented	server/test/sampleController.test.js — dismissal-only error updates save once and return full sample.
UT-UC5-07	UC5	Unit	Confirm uncertain error	Implemented	server/test/sampleController.test.js — confidence-only update preserves other fields.
UT-UC5-08	UC5	Unit	Valid new manual error	Implemented	server/test/sampleController.test.js — error index new appends educator-identified error.
UT-UC5-09	UC5	Unit	Invalid patch	Partially implemented	server/test/sampleController.test.js covers out-of-range index only; other validation branches are missing.
E2E-UC5-01	UC5	System E2E	Correct existing AI tags	Not implemented	No automated browser E2E test found.
E2E-UC5-02	UC5	System E2E	Restore removed tag	Not implemented	No automated browser E2E test found.
E2E-UC5-03	UC5	System E2E	Add an AI-missed error	Not implemented	No automated browser E2E test found.
E2E-UC5-04	UC5	System E2E	Rejected correction	Not implemented	No automated browser E2E test found.
RB-UC5-01	UC5	Robustness	Index validation	Partially implemented	sampleController.test.js covers out-of-range positive index only.
RB-UC5-02	UC5	Robustness	Patch-field validation	Not implemented	No matching automated test found.
RB-UC5-03	UC5	Robustness	Manual text validation	Not implemented	No matching automated test found.
RB-UC5-04	UC5	Robustness	Child-text integrity	Partially implemented	sampleController new-error test asserts a misspelling; broader whitespace/case integrity is not tested.
RB-UC5-05	UC5	Robustness	Save failure	Not implemented	No matching automated test found.
BI-UC5-01	UC5	Bottom-up integration	Valid existing-error patch	Partially implemented	sampleController.test.js covers dismissal/confidence with mocked Sample, not real MongoDB.
BI-UC5-02	UC5	Bottom-up integration	Valid new manual error	Partially implemented	sampleController.test.js covers controller with mock document.
BI-UC5-03	UC5	Bottom-up integration	Correction actions	Not implemented	No matching automated test found.
BI-UC5-04	UC5	Bottom-up integration	Invalid patch	Not implemented	No matching automated test found.
UT-UC6-01	UC6	Unit	Generate or Refresh recommendations	Not implemented	No matching automated test found.
UT-UC6-02	UC6	Unit	Active evidence exists	Implemented	server/test/generateReport.test.js — uses all analysed samples, excludes private identity/path data.
UT-UC6-03	UC6	Unit	No active errors	Implemented	server/test/generateReport.test.js — rejects students without analysed errors.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC6-04	UC6	Unit	Gemini recommendation mode	Implemented	server/test/recommendationEngine. test.js — grounding, mapping, validation and mock-selection tests.
UT-UC6-05	UC6	Unit	Generation failure	Implemented	server/test/recommendationEngine. test.js — retry, malformed JSON, timeout/size controls.
UT-UC6-06	UC6	Unit	Upsert succeeds	Implemented	server/test/recommendationRoutes. test.js — atomically upserts latest-only report.
UT-UC6-07	UC6	Unit	Student missing	Not implemented	No matching automated test found.
UT-UC6-08	UC6	Unit	Duplicate-key race	Implemented	server/test/recommendationRoutes. test.js — recovers when concurrent first upsert wins.
UT-UC6-09	UC6	Unit	Generation failure	Not implemented	No matching automated test found.
E2E-UC6-01	UC6	System E2E	Generate first recommendations	Not implemented	No automated browser E2E test found.
E2E-UC6-02	UC6	System E2E	Refresh outdated recommendations	Not implemented	No automated browser E2E test found.
E2E-UC6-03	UC6	System E2E	No active errors	Not implemented	No automated browser E2E test found.
E2E-UC6-04	UC6	System E2E	External generation failure	Not implemented	No automated browser E2E test found.
RB-UC6-01	UC6	Robustness	Missing/no-evidence input	Partially implemented	generateReport no-active-errors test exists; missing-student generation test missing.
RB-UC6-02	UC6	Robustness	External service failure	Partially implemented	Engine faults are tested; old-report persistence invariant is not.
RB-UC6-03	UC6	Robustness	Grounding integrity	Implemented	recommendationEngine. test.js evidence, worksheet and category validation tests.
RB-UC6-04	UC6	Robustness	Atomic replacement	Partially implemented	Atomic findOneAndUpdate options tested; real failure atomicity not integrated.
RB-UC6-05	UC6	Robustness	First-upsert race	Partially implemented	recommendationRoutes. test.js simulates E11000 recovery.
BI-UC6-01	UC6	Bottom-up integration	Upsert succeeds	Partially implemented	generateReport/ recommendationRoutes tests use mocked models; atomic options are asserted.
BI-UC6-02	UC6	Bottom-up integration	Generation failure	Not implemented	No matching automated test found.
BI-UC6-03	UC6	Bottom-up integration	Duplicate-key race	Partially implemented	recommendationRoutes. test.js simulates E11000 but not real concurrent database requests.
BI-UC6-04	UC6	Bottom-up integration	Active evidence exists	Not implemented	No matching automated test found.
BI-UC6-05	UC6	Bottom-up integration	Student missing / no active errors	Not implemented	No matching automated test found.
UT-UC7-01	UC7	Unit	Credentials valid	Not implemented	No matching automated test found.
UT-UC7-02	UC7	Unit	Credentials invalid	Not implemented	No matching automated test found.
UT-UC7-03	UC7	Unit	Credentials valid	Implemented	server/test/auth. integration.test.js — successful login returns username and email.
UT-UC7-04	UC7	Unit	Username or password missing	Implemented	server/test/auth. integration.test.js — missing password through error middleware.
UT-UC7-05	UC7	Unit	Account missing or password mismatch	Implemented	server/test/auth. integration.test.js — unknown username and wrong password tests.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC7-06	UC7	Unit	Client-side route gate	Partially implemented	client/test/RequireAuth.test.js covers redirect/allow; saveSession itself and LoginPage handoff are not tested.
E2E-UC7-01	UC7	System E2E	Log in successfully	Not implemented	No automated browser E2E test found.
E2E-UC7-02	UC7	System E2E	Reject invalid login	Not implemented	No automated browser E2E test found.
E2E-UC7-03	UC7	System E2E	Protected route gate	Not implemented	No automated browser E2E test found.
RB-UC7-01	UC7	Robustness	Credential validation	Implemented	auth.integration.test.js missing-password case.
RB-UC7-02	UC7	Robustness	Enumeration resistance	Implemented	auth.integration.test.js asserts same message in separate cases.
RB-UC7-03	UC7	Robustness	Response secrecy	Partially implemented	Successful integration test asserts exact username/email response; no broader client session payload test.
RB-UC7-04	UC7	Robustness	Client-only gate limitation	Not implemented	No matching automated test found.
BI-UC7-01	UC7	Bottom-up integration	Credentials valid	Partially implemented	auth.integration.test.js uses an in-memory model stub rather than isolated MongoDB.
BI-UC7-02	UC7	Bottom-up integration	Credentials invalid	Partially implemented	auth.integration.test.js covers both branches over HTTP with stubs.
BI-UC7-03	UC7	Bottom-up integration	Credentials valid	Partially implemented	auth.integration.test.js uses real HTTP route/controller but stubbed persistence.
BI-UC7-04	UC7	Bottom-up integration	Invalid login	Partially implemented	auth.integration.test.js covers these HTTP branches.
UT-UC8-01	UC8	Unit	Client validation passes	Not implemented	No matching automated test found.
UT-UC8-02	UC8	Unit	Client validation fails	Not implemented	No matching automated test found.
UT-UC8-03	UC8	Unit	Server password policy	Implemented	server/test/passwordPolicy.test.js — valid, every unmet rule, each single-rule failure and boundary tests.
UT-UC8-04	UC8	Unit	New password valid	Not implemented	No matching automated test found.
UT-UC8-05	UC8	Unit	Current password incorrect	Not implemented	No matching automated test found.
UT-UC8-06	UC8	Unit	Server password policy fails	Not implemented	No matching automated test found.
E2E-UC8-01	UC8	System E2E	Change password successfully	Not implemented	No automated browser E2E test found.
E2E-UC8-02	UC8	System E2E	Client rejects weak/mismatch	Not implemented	No automated browser E2E test found.
E2E-UC8-03	UC8	System E2E	Server rejects wrong current	Not implemented	No automated browser E2E test found.
RB-UC8-01	UC8	Robustness	Current-password secrecy	Not implemented	No matching automated test found.
RB-UC8-02	UC8	Robustness	Server-side strength enforcement	Partially implemented	Password policy is unit-tested; controller integration is missing.
RB-UC8-03	UC8	Robustness	Confirmation mismatch	Not implemented	No matching automated test found.
RB-UC8-04	UC8	Robustness	Persistence failure	Not implemented	No matching automated test found.
RB-UC8-05	UC8	Robustness	Boundary password	Partially implemented	Server policy boundary is tested; client form consistency and controller path are not.
BI-UC8-01	UC8	Bottom-up integration	New password valid	Not implemented	No matching automated test found.
BI-UC8-02	UC8	Bottom-up integration	Password validation fails	Not implemented	No matching automated test found.
BI-UC8-03	UC8	Bottom-up integration	New password valid	Not implemented	No matching automated test found.
BI-UC8-04	UC8	Bottom-up integration	Validation failures	Not implemented	No matching automated test found.
UT-UC9-01	UC9	Unit	Reset request accepted	Not implemented	No matching automated test found.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC9-02	UC9	Unit	Email error	Not implemented	No matching automated test found.
UT-UC9-03	UC9	Unit	Account exists	Implemented	server/test/auth. integration.test.js — sends reset email and stores token.
UT-UC9-04	UC9	Unit	Account missing	Implemented	server/test/auth. integration.test.js — identical response for unknown username and no email.
UT-UC9-05	UC9	Unit	Client validation fails	Not implemented	No matching automated test found.
UT-UC9-06	UC9	Unit	Token valid	Implemented	server/test/auth. integration.test.js — updates password and clears token on success.
UT-UC9-07	UC9	Unit	Server password policy fails	Implemented	server/test/auth. integration.test.js — weak password does not consume token.
UT-UC9-08	UC9	Unit	Token invalid or expired	Partially implemented	auth.integration.test.js name mentions expired token; fixture directly covers only an unknown token.
UT-UC9-09	UC9	Unit	Resend accepted or error	Not implemented	No matching automated test found.
E2E-UC9-01	UC9	System E2E	Forgot and reset successfully	Not implemented	No automated browser E2E test found.
E2E-UC9-02	UC9	System E2E	Unknown username privacy	Not implemented	No automated browser E2E test found.
E2E-UC9-03	UC9	System E2E	Reject invalid/expired link	Not implemented	No automated browser E2E test found.
E2E-UC9-04	UC9	System E2E	Reject weak/mismatch locally	Not implemented	No automated browser E2E test found.
RB-UC9-01	UC9	Robustness	Username privacy	Implemented	auth.integration.test.js privacy-pair tests.
RB-UC9-02	UC9	Robustness	Email provider failure	Not implemented	No matching automated test found.
RB-UC9-03	UC9	Robustness	Weak password	Implemented	auth.integration.test.js weak password does not consume token.
RB-UC9-04	UC9	Robustness	Invalid/expired/reused token	Partially implemented	Unknown/expired is tested; explicit token reuse after success is not.
RB-UC9-05	UC9	Robustness	Successful token consumption	Implemented	auth.integration.test.js success case.
BI-UC9-01	UC9	Bottom-up integration	Account exists	Partially implemented	auth.integration.test.js uses in-memory account records and mocked email.
BI-UC9-02	UC9	Bottom-up integration	Token valid	Partially implemented	auth.integration.test.js covers save outcome with stubs.
BI-UC9-03	UC9	Bottom-up integration	Account exists/missing	Partially implemented	auth.integration.test.js covers the privacy pair with mocked Account and email.
BI-UC9-04	UC9	Bottom-up integration	Reset branches	Partially implemented	auth.integration.test.js covers weak, invalid/expired and success branches with stubs.

Coverage summary by use case and level

Use Case	Level	Implemented	Partial	Missing	Conflict	Total
UC1	Unit	3	0	5	0	8
UC1	Bottom-up integration	0	2	2	0	4
UC1	System E2E	0	0	3	0	3
UC1	Robustness	2	1	2	0	5
UC2	Unit	3	0	4	0	7
UC2	Bottom-up integration	0	0	4	0	4
UC2	System E2E	0	0	2	0	2
UC2	Robustness	1	2	1	0	4
UC3	Unit	1	2	3	0	6
UC3	Bottom-up integration	0	2	2	0	4
UC3	System E2E	0	0	3	0	3
UC3	Robustness	2	1	1	0	4
UC4	Unit	4	2	1	0	7
UC4	Bottom-up integration	0	2	2	0	4
UC4	System E2E	0	0	3	0	3
UC4	Robustness	2	2	0	0	4
UC5	Unit	7	1	1	0	9
UC5	Bottom-up integration	0	2	2	0	4
UC5	System E2E	0	0	4	0	4
UC5	Robustness	0	2	3	0	5
UC6	Unit	6	0	3	0	9
UC6	Bottom-up integration	0	2	3	0	5
UC6	System E2E	0	0	4	0	4
UC6	Robustness	1	4	0	0	5
UC7	Unit	3	1	2	0	6
UC7	Bottom-up integration	0	4	0	0	4
UC7	System E2E	0	0	3	0	3
UC7	Robustness	2	1	1	0	4
UC8	Unit	1	0	5	0	6
UC8	Bottom-up integration	0	0	4	0	4
UC8	System E2E	0	0	3	0	3
UC8	Robustness	0	2	3	0	5

Use Case	Level	Implemented	Partial	Missing	Conflict	Total
UC9	Unit	4	1	4	0	9
UC9	Bottom-up integration	0	4	0	0	4
UC9	System E2E	0	0	4	0	4
UC9	Robustness	3	1	1	0	5

Key conclusions

- Strong coverage exists for AI validation, recommendation safety, trend functions, authentication/reset server branches and UC5 embedded-error updates.
- Main gaps are browser E2E, upload/polling UI, report-read controllers, recommendation UI, change-password flow and forgot/reset client pages.
- Existing files named integration commonly use model stubs, so they are Partial against an isolated-database boundary.
- No automated test conflict with the corrected sequence diagrams was found.